1 Background: X-Ray Scattering and the Debye Equation

When a beam of X-rays is fired at a molecule, each atom deflects ("scatters") the beam by an amount that depends on the atom's identity and the deflection angle q. Each atom has a characteristic *atomic form factor* $f_k(q)$ encoding how strongly it scatters at angle q. By measuring the total scattered intensity across all angles, we obtain a *scattering profile* $I(q)$ which is a unique fingerprint of the molecule's 3-D shape.

The exact formula, the **Debye equation**, sums the pairwise interaction of every atom with every other atom:

$$I(q) = \sum_{\alpha} \sum_{\beta} f_\alpha(q)\, f^*_\beta(q)\ \operatorname{sinc}\big(q\,|\mathbf{r}_\alpha - \mathbf{r}_\beta|\big) \tag{1}$$

where $\operatorname{sinc}(x) = \sin(x)/x$ captures the intuition that atoms far apart interfere less strongly. Because every pair must be evaluated, this scales as $O(N^2)$ in the number of atoms. **CRYSOL ?** approximates Eq. (1) by decomposing the molecular shape into a sum of angular "modes" (spherical harmonics), truncated at order L:

$$I(q) = \sum_{l=0}^{L} \sum_{m=-l}^{l} \big|A_{lm}(q) - \rho_0\, C_{lm}(q) + \delta\rho\, B_{lm}(q)\big|^2 \tag{2}$$

where A_{lm}, C_{lm}, B_{lm} encode contributions from the molecule itself, the displaced solvent, and the surrounding hydration shell, respectively. This reduces the cost to $O(N \cdot L^2)$, with $L \leq 20$ sufficient for most proteins. A live demo is available at: (ADD DEMO LINK HERE).

This raises the central question of this project: *can a neural network learn to predict $I(q)$ directly from the 3-D atomic coordinates of a molecule?*

2 Proposed Architecture

The ground truth for each molecule is computed via the **Stuhrmann decomposition** (in-vacuo, no solvent corrections):

$$B_{lm}(q) = \sum_i f_i(q)\, j_l(qr_i)\, Y_{lm}(\theta_i, \phi_i), \qquad I(q) = (4\pi)^2 \sum_{l=0}^{L} \sum_{m=-l}^{l} |B_{lm}(q)|^2 \tag{3}$$

where j_l are spherical Bessel functions and Y_{lm} are spherical harmonics. The only label available per molecule is the intensity vector $I(q) \in \mathbb{R}^{|Q|}$.

The individual $|B_{lm}(q)|^2$ terms are not directly supervised, so the network could in principle output $I(q)$ directly. However, we instead propose to predict the $\frac{(L+1)(L+2)}{2} \times |Q|$ matrix of non-negative scalars $|B_{lm}(q)|^2$ and sum them to produce $\hat{I}(q)$, with an MSE loss on the final sum. This mirrors the structure of the ground-truth computation and provides useful inductive bias: each (l, m) head specialises in a single angular frequency of the scattering pattern, and non-negativity is trivially enforced (outputs are squared). The coefficients B_{lm} are rotationally variant, but $|B_{lm}|^2$ is rotationally invariant, so no equivariant machinery is required. The full architecture is as follows:

1. **TODO**

3 Data Processing

1.3 million molecules (around 50GB) were processed into a database, spanning small organic molecules to large metal organic frameworks. These were pulled from the following databases: QM9; tmQF; viro3D; rcsb (filtered <9k atoms); cambridge energy landscape; and QMML. Full source attributions can be found here (add link). The datapipeline was as followed: 1. data was bulk downloaded 2. scripts converted molecular formats such as .cif/.pdb to cartesian .xyz format 3. nested frames were extracted into seperate files 4. data was scanned for wrong and/or dummy sites (ie: placeholders), and they were removed from the files 5. deuterium (D) was translated to hydrogen 5. ionic makeup was tallied (can be found here) 6. data was fed to python scripts to calculate I(q) ground truths and store data as .hdf5 file.